

- 6 (a) A simple a.c. generator which consists of a rectangular coil with 40 turns is rotating at 50 revolutions per second in a uniform magnetic field of flux density 50 mT. The coil is 50 cm long and 35 cm wide, as shown in Fig. 6.1.

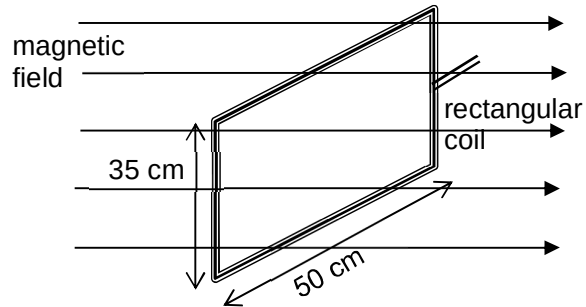


Fig. 6.1

- (i) Define *magnetic flux*.

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[1]

The magnetic flux through a **plane surface** is defined as the **product of the magnetic flux density normal to the surface and the area of the surface**. **B1**

- (ii) Calculate the maximum e.m.f. induced in the coil.

maximum e.m.f. = V [2]

$$E = NBA\omega$$

$$= (40)(50 \times 10^{-3})(50 \times 10^{-2})(35 \times 10^{-2})(2\pi)(50)$$

$$= 109.96 = 110 \text{ V (2 s.f.)}$$

C1

A1

- (b) The output from the generator in part (a) is connected to the input of a transformer. The transformer has 40 turns on its primary coil and 10 turns on its secondary coil. The transformer may be considered to be ideal. The output from the transformer is connected to a resistor of resistance 100Ω and a diode, as shown in Fig. 6.2.

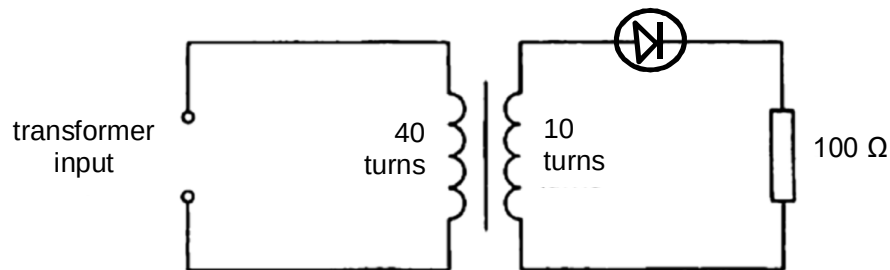


Fig. 6.2

Calculate the root-mean-square voltage across the resistor.

root-mean-square voltage = V [3]

Let V_0 be the peak voltage across the resistor.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

M1

$$\frac{V_0}{110} = \frac{10}{40}$$

$$V_0 = 27.5 \text{ V (2 s.f.)}$$

Since the voltage across the resistor is half-wave rectified,
Mean of v_0^2 for a half-wave rectified voltage
= area under the graph/period
= $(v_0^2 \times T)/(4 \times T)$
= $v_0^2/4$

Hence the root-mean-square, $V_{\text{rms}} = V_0/2$
= 14 V (2 s.f.)

M1

A1

- (c) In an attempt to draw the circuit diagram that represents the system in part (b), a student drew Fig. 6.3 instead.

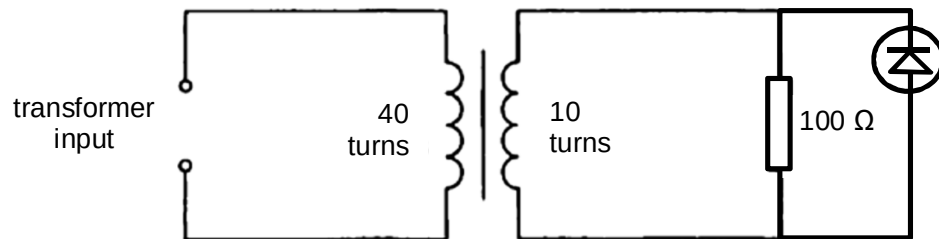


Fig. 6.3

Suggest, with a reason, what will happen if the circuit is set up according to Fig. 6.3.

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[2]

When the e.m.f. set up in the secondary circuit is such that the **diode is in forward bias, all / most current will by-pass the resistor and flow only via the diode back to the secondary coil.** B1

This means that the circuit will be short-circuited and the system could be damaged by **overheating.** B1